FORMULAE LIST

Circle

$x^2 + y^2 + 2gx + 2fy + c = 0$ represents a circle centre $(-g, -f)$, radius $\sqrt{g^2 + f^2 - c}$.

$(x - a)^2 + (y - b)^2 = r^2$ represents a circle centre (a, b) and radius r.

Scalar product

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta, \text{ where } \theta \text{ is the angle between } \mathbf{a} \text{ and } \mathbf{b}$$

or

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 \text{ where } \mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

Trigonometric formulae

$$\begin{aligned}
\sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \\
\cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B \\
\sin 2A &= 2 \sin A \cos A \\
\cos 2A &= \cos^2 A - \sin^2 A \\
&= 2\cos^2 A - 1 \\
&= 1 - 2\sin^2 A
\end{aligned}$$

Table of standard derivatives

$f(x)$	$f'(x)$
$\sin ax$	$a \cos ax$
$\cos ax$	$-a \sin ax$

Table of standard integrals

$f(x)$	$\int f(x)\, \mathrm{d}x$
$\sin ax$	$-\frac{1}{a} \cos ax + C$
$\cos ax$	$\frac{1}{a} \sin ax + C$

1. Find $\int 2\cos\left(3x - \frac{\pi}{4}\right)\,\mathrm{d}x$. (2)

2. Solve $\log_2(9-x) + \log_2 x = 3$, where $0 < x < 9$. (4)

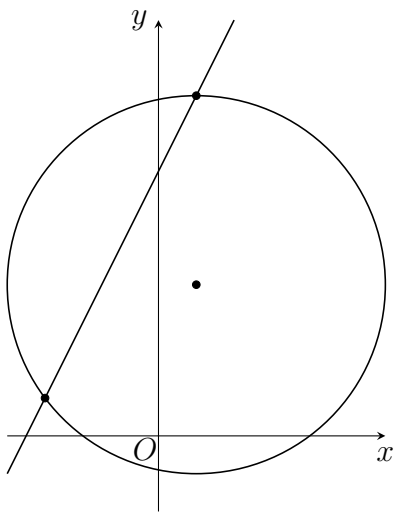
3. (a) Express $5\cos x^\circ - 3\sin x^\circ$ in the form $k\cos(x+a)^\circ$, $k > 0$, $0 < a < 360$. (4)

   (b) Hence state:

      i. The minimum value of the function $f(x) = 5\cos x^\circ - 3\sin x^\circ$. (1)

      ii. The value of x for which it occurs where $0 < x < 360$. (1)

4. Find the coordinates of the points where the line with equation $y = 2x + 3$ intersects the circle with equation $x^2 + y^2 - 2x - 8y - 8 = 0$. (4)



5. Determine the gradient of the tangent to the the curve with equation $y = \frac{1}{\sqrt{4x-8}}$ at the point where $x = 6$. (4)

6. Find $\int_1^5 \left(-6(2x-1)^{-2}\right)\,\mathrm{d}x$. (4)

7. Researchers are studying the growth of a type of bacteria in certain conditions. The number of bacteria B present after t hours may be modelled by the equation:

$$B = B_0 e^{kt}$$

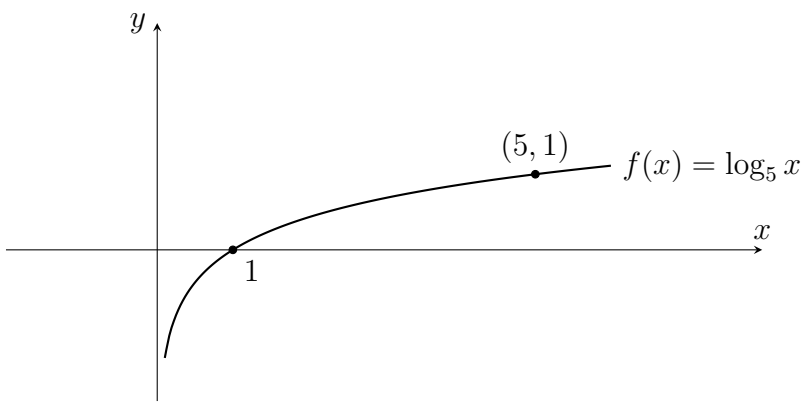
Where B_0 is the initial number of bacteria present.

After 7 hours, the number of bacteria present has increased from 250 to 1600.

(a) Determine the value of k to three significant figures. (3)

(b) Find the number of bacteria present after one **day**, to the nearest 10. (2)

8. The diagram below shows the graph the function $f(x) = \log_5 x$, where $x > 0$. (2)



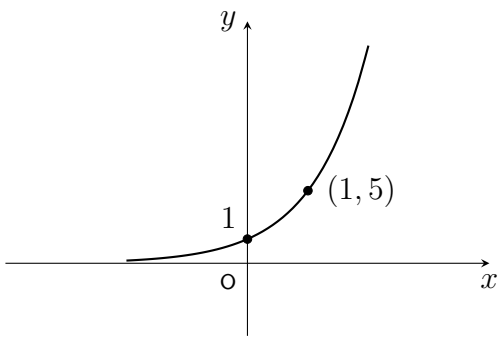
The inverse function, f^{-1}, exists.
Sketch the graph of the inverse function.

9. A circle C has equation $x^2 + y^2 + 2x + 4y - 27 = 0$. (3)
The point P$(3, 2)$ lies on the circle C.
Determine the gradient of the tangent at P.

END OF QUESTIONS

ANSWERS - Practice Exam E

1. $\frac{2}{3}\sin\left(3x - \frac{\pi}{4}\right) + \mathrm{C}$
2. $x = 1, x = 8$
3. (a) $\sqrt{34}\cos(x + 31.0)^\circ$
   (b) i. min $= -\sqrt{34}$
   ii. Occurs when $x = 149.0$
4. $(-3, 1)$ and $(1, 9)$
5. $-\frac{1}{32}$
6. $-\frac{8}{3}$
7. (a) $k = 0.265$
   (b) 7820
8. Sketch:



9. -1